

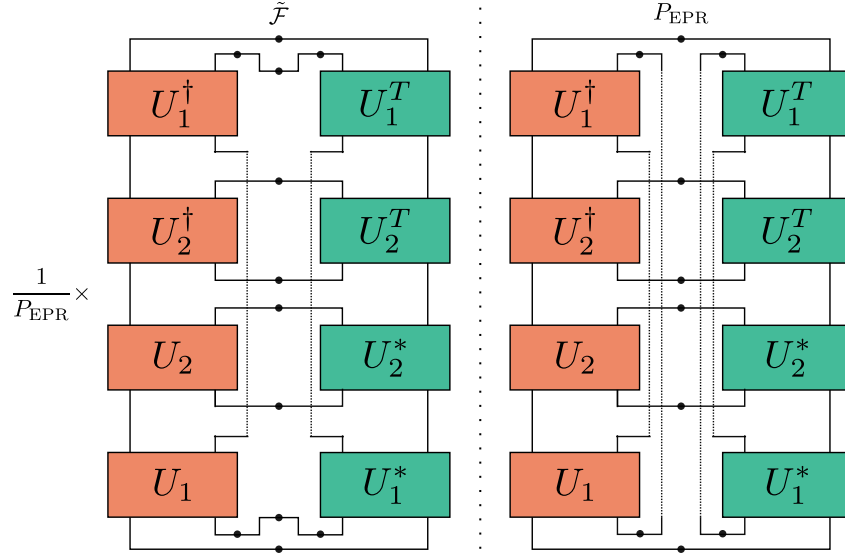


Figure 16: The fidelity of the second Hayden-Preskill-like protocol of the main text. The numerator trivially contracts to yield a constant factor and the non-trivial dependence on the unitary appears in the probability of measuring a bell pair, P_{EPR}

C Details of HP fidelity and derivation of Eq. (15)

The fidelity for Charlie’s decoding protocol is shown diagrammatically in Figure 16 where the only non-trivial factor is given by the probability of measuring a Bell pair, P_{EPR} . By comparing the diagram for P_{EPR} and that for the original decoder’s fidelity, $\mathcal{F}$ in Figure 15, it is apparent that

$$P_{\text{EPR}} \equiv \mathcal{F}. \quad (22)$$

The numerator in $\tilde{\mathcal{F}}$ trivially contracts to yield a factor of d_I^{-2} where d_I is the Hilbert space dimension of the impurity. Altogether we have the following relation between the two fidelities:

$$\mathcal{F} = \frac{1}{\tilde{\mathcal{F}} d_I^2}. \quad (23)$$

Substituting this into Eq. (11) and making use of Eq. (14) (along with $d_I = 2$) then yields Eq. (15).